BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

INTRODUCTION TO WEB PROGRAMMING

Unit 3

Tables and Text Styling in HTML

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1. INTRODUCTION

In the field of web development, the ability to craft well-structured and visually appealing content is supreme. Tables and text styling stand as two fundamental tools in the collection of any web designer or developer seeking to create engaging and organized web pages. In this chapter, we delve into the art of utilizing HTML to construct tables that neatly present data and information, providing both form and function to your web content. Simultaneously, we explore the world of text styling, leveraging CSS (Cascading Style Sheets) to breathe life into your text elements, adding flair, readability, and personality to your web pages.

Tables, as the backbone of tabular data presentation, allow you to arrange information in rows and columns, making it comprehensible and structured. Understanding how to create and manipulate tables is essential for anyone dealing with data representation, from simple charts to intricate data-driven applications. We'll walk you through the HTML elements required to construct tables, discuss best practices for accessibility, and demonstrate how to style tables to match your website's aesthetic.

On the other side of the spectrum, text styling empowers you to transform plain text into captivating content. With CSS, you can control the font, size, color, alignment, and more, turning ordinary words into visually striking headlines, paragraphs, and calls to action. The ability to style text is the key to creating engaging user interfaces and making your content stand out in the crowded digital landscape. In this chapter, we'll guide you through various text styling techniques, from the basics of choosing fonts and adjusting sizes to more advanced concepts like shadows, text decorations, and responsive typography.

This chapter will equip you with the knowledge and techniques needed to harness the power of tables and text styling in HTML.

1.1 Learning Objectives

At the end of this topic, the students will be able to:

- *Identify the essential attributes required for table construction, such as border, headers, rows, and cells.*
- *Summarize the key principles of creating well-structured tables that enhance user experience.*
- *Implement CSS styles to format and customize table elements, such as changing border styles, cell spacing, and cell padding.*
- *Analyze a web page's design choices, including font selection, size, color, and alignment, and assess their impact on user engagement.*
- *Develop a responsive web page that showcases advanced text styling techniques and demonstrates creativity in text presentation while adhering to best practices.*

2. CREATING AND USING TABLES IN HTML

An HTML table is a structured grid of data or information that consists of rows and columns. Tables are a fundamental part of web development and are commonly used to organize and display data in a structured format. They allow you to present information in a visually organized manner, making it easier for users to understand and navigate the content.

Here's a detailed explanation of HTML tables along with an example:

Basic Structure of an HTML Table:

1. An HTML table is created using a combination of HTML elements, mainly the **<table>**, **<tr>**, **<th>**, and **<td>** elements.

Tag	Description
<table>	Defines a table
<th>	Defines a header cell in a table
<tr>	Defines a row in a table
<td>	Defines a cell in a table
<caption>	Defines a table caption
<colgroup>	Specifies a group of one or more columns in a table for formatting
<col>	Specifies column properties for each column within a <colgroup> element
<thead>	Groups the header content in a table
<tbody>	Groups the body content in a table
<tfoot>	Groups the footer content in a table

Fig 3. 1

The HTML table is created using a combination of HTML elements that define the structure of the table and its contents. Here's the syntax of an HTML table along with an explanation of each element:

Syntax:

```
<table>
<caption>Table Caption</caption>
<thead>
<tr>
<th>Header 1</th>
<th>Header 2</th>
<!-- Add more header cells if needed -->
</tr>
</thead>
<tbody>
<tr>
<td>Data 1, Row 1</td>
<td>Data 2, Row 1</td>
<!-- Add more data cells for this row -->
</tr>
<tr>
<td>Data 1, Row 2</td>
<td>Data 2, Row 2</td>
<!-- Add more data cells for this row -->
</tr>
<!-- Add more rows with data as needed -->
</tbody>
<tfoot>
<tr>
<td>Footer 1</td>
<td>Footer 2</td>
<!-- Add more footer cells if needed -->
</tr>
</tfoot>
</table>
```

Now, let's get into each part of the HTML table syntax:

1. **<table>**: This is the container element for the entire table. It defines the start and end of the table.
2. **<caption>** (optional): This element is used to provide a title or caption for the table. It appears above the table and provides context for the table's content.
3. **<thead>**: This element represents the table's header section. It is used to group the header row(s) of the table.
4. **<tr>**: Stands for "table row." It is used to define a row within the table, whether it's in the header, body, or footer section.

5. **<th>**: Stands for "table header." It is used to define header cells within a table row. Header cells are typically used in the first row or rows to label or describe the content of the columns. They are displayed in bold by default.
6. **<tbody>** (optional): This element represents the table's body section. It is used to group the data rows of the table.
7. **<td>**: Stands for "table data." It is used to define regular data cells within a table row. These cells contain the actual data or information.
8. **<tfoot>** (optional): This element represents the table's footer section. It is used to group footer row(s) of the table, typically used for summarizing data or providing additional information at the end of the table.

Important Notes:

- The **<thead>**, **<tbody>**, and **<tfoot>** sections are not required, but they are useful for structuring your table and can improve accessibility.
- You can have multiple rows of header cells or data cells as needed within the **<thead>**, **<tbody>**, or **<tfoot>** sections.
- The **colspan** and **rowspan** attributes can be used to span cells across multiple columns or rows, respectively.

This syntax provides a flexible way to create tables in HTML, allowing you to organize and present data in a structured manner. Depending on your specific needs, you can customize the table structure by adding or omitting sections and cells. Additionally, you can style the table using CSS to control its appearance on your web page.

Top of Form

Example :

```
<!DOCTYPE html>
<html>
<head>
   <title>Sample HTML Table</title>
</head>
<body>
   <h1>Sample HTML Table</h1>
      <!-- Start of the table -->
   <table border="1">
      <!-- Header row -->
      <tr>
         <th>Header 1</th>
         <th>Header 2</th>
         <th>Header 3</th>
      </tr>
            <!-- First data row -->
      <tr>
         <td>Data 1, Row 1</td>
         <td>Data 2, Row 1</td>
         <td>Data 3, Row 1</td>
      </tr>
            <!-- Second data row -->
      <tr>
         <td>Data 1, Row 2</td>
         <td>Data 2, Row 2</td>
         <td>Data 3, Row 2</td>
      </tr>
            <!-- Third data row -->
      <tr>
         <td>Data 1, Row 3</td>
         <td>Data 2, Row 3</td>
         <td>Data 3, Row 3</td>
      </tr>
   </table>
   <!-- End of the table -->
</body>
</html>
```

1. **Document Structure:**
   - The HTML document is structured with an opening <**html**> tag, followed by a <head> section, and then a <body> section.
2. **Page Title:**
   - The title of the web page is set using the <t**itle**> element. This title appears in the browser's title bar or tab.
3. **Heading:**
   - An <**h1**> heading element is used to provide the main title for the web page. It is displayed as the largest and most prominent text.
4. **HTML Table:**
   - The <**table**> element defines the start of the HTML table. It's the container for the entire table.
5. **Table Structure:**
   - The table structure consists of rows and cells. Rows are defined using the <tr> element, and cells are defined using either <**th**> for header cells or <td> for data cells.
   - The first row is typically the header row, and it contains <**th**> cells to label the columns.
   - Subsequent rows are data rows, containing <**td**> cells with actual data.
   - This part defines the header row of the table using the <**tr**> element, which represents a table row. Inside the header row, we have three header cells (<**th**>) labeled "Header 1," "Header 2," and "Header 3." Header cells are typically used to label or describe the content of the columns.
   - Following the header row, we have the first data row. It consists of three data cells (<**td**>), each containing data for the respective column. In this case, the data cells hold "Data 1, Row 1," "Data 2, Row 1," and "Data 3, Row 1."
   - There are two more data rows (<**tr**>) following this pattern, each containing data cells with unique content for their respective columns.
6. **Table Closing Tag:**
   - The <**/table**> tag marks the end of the HTML table definition.

7. **Closing Tags:**
   - Finally, the **</body>** and **</html>** tags are used to close the HTML document.

In summary, the HTML table structure consists of a **<table>** element as the container, with rows defined by **<tr>** elements, and cells within rows defined as either **<th>** (header cells) or **<td>** (data cells). The header row labels the columns, and the data rows contain the actual information. The structure is enclosed within an HTML document with a defined title and heading.

The output of the above code will be

Sample HTML Table

Header 1	Header 2	Header 3
Data 1, Row 1	Data 2, Row 1	Data 3, Row 1
Data 1, Row 2	Data 2, Row 2	Data 3, Row 2
Data 1, Row 3	Data 2, Row 3	Data 3, Row 3

Fig 3. 2

2.2 HTML Table Borders

To add a border to an HTML table, you can use the border attribute on the <table> element. Here's how you can do it:

Syntax :

```
<table border="1">
  <!-- Table content goes here -->
</table>
```

In the syntax above, we've set the border attribute to "1." This value specifies the width of the border around the table. A value of "1" creates a thin border. You can adjust this value to make the border thicker if desired.

Here's a breakdown:

<table>: This is the opening tag for the table element.

border="1": This is the border attribute with a value of "1," indicating a 1-pixel border around the table.

It's important to note that using the border attribute directly in your HTML code is a simple way to add borders for demonstration purposes. In practice, for more control and better styling, it's recommended to use CSS (Cascading Style Sheets) to define table borders and styles. CSS provides more flexibility in terms of border thickness, style, and color.

Example :

```
<!DOCTYPE html>
<html>
<head>
   <title>Sample HTML Table</title>
</head>
<body>
   <h1>Sample HTML Table</h1>
   <!-- Start of the table -->
   <table border="10  ">
      <!-- Header row -->
      <tr>
         <th>Header 1</th>
         <th>Header 2</th>
            </tr>
            <!-- First data row -->
      <tr>
         <td>Data 1, Row 1</td>
         <td>Data 2, Row 1</td>
             </tr>
        </table>
   <!-- End of the table -->
</body>
</html>
```

The output of this example will be :

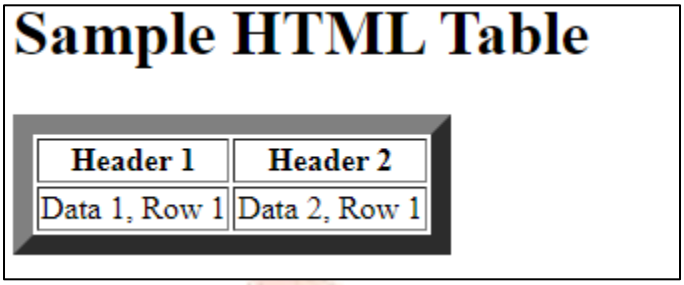

Fig 3. 3

2.3 Styling a HTML table

You can style an HTML table without using an external CSS file by applying inline styles directly to the HTML elements. While it's generally recommended to use CSS for styling to separate content from presentation, inline styles are useful for quick and simple adjustments. Here's an example of how to style an HTML table without CSS:

```
<table border="1" cellpadding="10" cellspacing="0" style="border-collapse:
collapse; width: 100%; text-align: center;">
   <!-- Header row -->
   <tr style="background-color: #333; color: #fff; font-weight: bold;">
      <th>Header 1</th>
      <th>Header 2</th>
      <th>Header 3</th>
   </tr>

   <!-- First data row -->
   <tr style="background-color: #f2f2f2;">
      <td>Data 1, Row 1</td>
      <td>Data 2, Row 1</td>
      <td>Data 3, Row 1</td>
   </tr>

   <!-- Second data row -->
   <tr style="background-color: #fff;">
      <td>Data 1, Row 2</td>
      <td>Data 2, Row 2</td>
      <td>Data 3, Row 2</td>
   </tr>
</table>
```

In this example:

- We use the style attribute within the HTML elements to apply inline styles directly.
- **border="1"** sets a border width of 1 pixel around the entire table.
- **cellpadding="10"** adds 10 pixels of padding within each cell.
- **cellspacing="0"** specifies no spacing between cells.
- **border-collapse:** collapse; collapses the borders of adjacent cells into a single border line.
- **width: 100%;** makes the table span the full width of its container.
- **text-align: center**; centers the text within the cells.

For individual rows (<**tr**>), we set background colors, text color, and font weight to style the header row and alternating data rows differently.

Lets get into the details of it :

<table border="1" cellpadding="10" cellspacing="0" style="border-collapse: collapse; width: 100%; text-align: center;">

- **border="1":** This attribute sets the border width of the table to 1 pixel, creating a visible border around the entire table.
- **cellpadding="10":** This attribute adds 10 pixels of padding within each cell. It creates space between the cell content and the cell's border.
- **cellspacing="0":** This attribute specifies that there should be no spacing between cells. Setting it to "0" ensures that cells are adjacent to each other without any gaps.

style="border-collapse: collapse; width: 100%; text-align: center;":

- **border-collapse: collapse;:** This inline style rule collapses the borders of adjacent cells into a single border line. It creates a cleaner and more visually appealing appearance for the table.
- **width: 100%;:** This inline style rule sets the width of the table to 100% of its containing element, which typically means the table will span the full width of its container. This makes the table responsive to the width of the screen or containing element.

- **text-align: center;:** This style rule centers the text content horizontally within each cell. All text within the table, including both header cells (<**th**>) and data cells (<**td**>), will be centered.

Moving on to the individual rows:

<!-- Header row --> <tr style="background-color: #333; color: #fff; font-weight: bold;">

- **style="background-color: #333; color: #fff; font-weight: bold;":**
- **background-color: #333;:** This style rule sets the background color of the header row to a dark gray color (#333).
- **color: #fff;:** This rule sets the text color of the header row to white (#fff) for better contrast against the dark background.
- **font-weight: bold;:** This rule makes the text in the header row bold for emphasis.

Similarly, the style is applied to the first and second data rows, but with different background colors to create a striped effect:

<!-- First data row --> <tr style="background-color: #f2f2f2;">

<!-- Second data row --> <tr style="background-color: #fff;">

- **background-color** is used to set the background colors of these rows to light gray (#f2f2f2) for the first data row and white (#fff) for the second data row, creating a visual distinction between them.

These inline styles allow you to customize the appearance of the table and its elements directly within the HTML code. However, for more extensive styling or consistent styles across multiple pages, using an external CSS file is typically preferred for maintainability and organization.

3. FONT SIZE AND ATTRIBUTES IN HTML

In HTML, you can set font size and attributes for text elements without using CSS by using HTML attributes and inline styles. Here's a detailed explanation along with syntax and examples:

1. *Font Size (`size` Attribute):*

You can set the font size directly on specific HTML elements using the `size` attribute. This attribute is available on some HTML elements such as headings (`<h1>`, `<h2>`, ... `<h6>`), paragraph elements (`<p>`), and font elements (`<font>`).

Syntax:

```
<element size="value">Text content</element>
```

- `<element>`: The HTML element you want to apply the font size to (e.g., `<h1>`, `<p>`).
- `size="value"`: The `size` attribute where you specify the font size. The value can be a number, which represents the size, or a relative size such as "small," "medium," or "large."

Example:

```
<h1 size="6">This is a large heading</h1>
<p size="3">This is a medium-sized paragraph.</p>
<font size="2">This is a small font.</font>
```

In this example, we've applied different font sizes to a heading, paragraph, and font element using the `size` attribute.

2. *Font Attributes (Deprecated):*

HTML used to support font-related attributes like `face`, `color`, and `size` directly on text elements. However, these attributes are deprecated in HTML5 and are no longer recommended for use. Instead, CSS should be used for styling text and fonts.

Syntax (Deprecated - Not Recommended):

```
<element face="font-family" color="text-color" size="font-size">Text content</element>
```

- `<element>`: The HTML element.
- `face="font-family"`: Specifies the font family (e.g., Arial, Times New Roman).

- `color="text-color"`: Defines the text color (e.g., "red," "#333333").
- `size="font-size"`: Sets the font size (e.g., "2," "small," "medium").

Example (Deprecated - Not Recommended):

```
<p face="Arial" color="blue" size="3">This is a blue, size 3, Arial font paragraph.</p>
```

While these attributes were commonly used in older versions of HTML, it's important to note that they are considered outdated in modern web development, and CSS is the recommended way to style text and fonts. For best practices, it's advisable to use CSS for styling whenever possible.

4. LIST TYPES AND ITS TAGS

An HTML list is a way to organize and present information in a structured, ordered, or unordered format on a web page. Lists are useful for displaying items, creating navigation menus, outlining content hierarchies, and more. There are three primary types of HTML lists: unordered lists (`<ul>`), ordered lists (`<ol>`), and definition lists (`<dl>`). Let's explore each type in detail with syntax and examples:

Tag	Description
<ul>	Defines an unordered list
<ol>	Defines an ordered list
<li>	Defines a list item
<dl>	Defines a description list
<dt>	Defines a term in a description list
<dd>	Describes the term in a description list

Fig 3. 4

1. *Unordered List (`<ul>`):*

An unordered list presents items with bullet points or other markers. Each list item is enclosed in `<li>` (list item) tags.

Syntax:

```
<ul>
   <li>Item 1</li>
   <li>Item 2</li>
   <li>Item 3</li>
</ul>
```

Example:

```
<ul>
   <li>Apples</li>
   <li>Oranges</li>
   <li>Bananas</li>
</ul>
```

The output will be

An unordered HTML list

- Apples
- Oranges
- Bananas

Fig 3. 5

2. *Ordered List (`<ol>`):*

An ordered list presents items with sequential numbers or letters. Each list item is enclosed in `<li>` tags.

Syntax:

```
<ol>
   <li>Item 1</li>
   <li>Item 2</li>
```

```
    <li>Item 3</li>
  </ol>
```

Example:

```
  <ol>
    <li>First</li>
    <li>Second</li>
    <li>Third</li>
  </ol>
```

The output will be

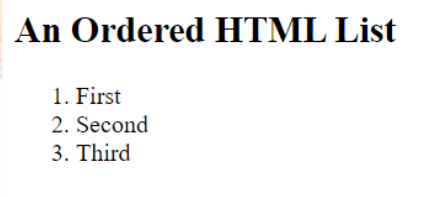

Fig 3. 6

3. *Definition List (`<dl>`):*

A definition list pairs terms (defined terms) with their corresponding definitions. Each term is enclosed in a `<dt>` (definition term) tag, and each definition is enclosed in a `<dd>` (definition description) tag.

Syntax:

```
  <dl>
    <dt>Term 1</dt>
    <dd>Definition 1</dd>
    <dt>Term 2</dt>
    <dd>Definition 2</dd>
  </dl>
```

Example:

```
  <dl>
    <dt>HTML</dt>
    <dd>Hypertext Markup Language</dd>
```

```
    <dt>CSS</dt>
    <dd>Cascading Style Sheets</dd>
  </dl>
```

The output will be

An unordered HTML list

HTML
    Hypertext Markup Language
CSS
    Cascading Style Sheets

Common Attributes for Lists:

- `type` (for `<ol>`): Specifies the type of numbering or bullet points (e.g., `type="1"` for numeric, `type="A"` for uppercase letters).
- `start` (for `<ol>`): Specifies the starting number for ordered lists.
- `reversed` (for `<ol>`): Reverses the counting order in ordered lists.

Lists can be nested within one another to create complex structures, and you can apply CSS styles to customize their appearance, including the shape and style of bullets or numbering, indentation, spacing, and more. Lists are fundamental for organizing content and enhancing the readability and structure of web pages.

5. SUMMARY

The chapter on HTML tables and lists covers essential HTML elements and structures for organizing and presenting content on web pages. It discusses the creation of tables to display data in rows and columns, as well as the use of unordered, ordered, and definition lists to organize information. These elements enable web developers to create structured and accessible content, enhancing both readability and user experience. Additionally, the chapter emphasizes the importance of semantic mark-up and accessibility considerations, ensuring that web content is not only visually appealing but also compliant with web standards and accessible to all users.

6. QUESTIONS

Self-Assessment questions

SELF-ASSESSMENT QUESTIONS – 1

1. What HTML element defines a table?
2. What HTML element is used for header cells in a table?
3. How can you make table borders collapse into a single line?
4. What attribute specifies the space between table cells?
5. What HTML element defines a table row?
6. What HTML element creates an unordered list?
7. What tag represents a list item in HTML?
8. What attribute in an ordered list changes the numbering style to Roman numerals?
9. Which HTML element is used to define a description list?
10. What tag defines a term in a definition list?

Terminal questions

1. Explain the structure of an HTML table and provide an example of when and why you would use one in web development.
2. Describe the key attributes used in HTML tables and their respective purposes. How can these attributes be used to control the table's appearance and behavior?
3. Discuss the importance of accessibility when creating HTML tables. What are some best practices for ensuring that tables are accessible to individuals with disabilities, and why is this important in web development?
4. Compare and contrast unordered lists (<**ul**>), ordered lists (<**ol**>), and definition lists (<**dl**>) in HTML. Provide examples of scenarios where each type of list would be most appropriate.
5. Explain the concept of list nesting in HTML. How can you nest lists within each other, and what are some practical use cases for nested lists in web design and content organization?
6. Describe the role of semantic markup in lists and its significance for search engine optimization (SEO) and web accessibility. How does using semantic HTML tags impact the way lists are interpreted by browsers and assistive technologies?

7. ANSWERS

Self-Assessment questions

1. table
2. th
3. border-collapse
4. cell spacing
5. tr
6. ul
7. li
8. type
9. dl
10. dt

Terminal question

1. Refer section 2
2. Refer section 2
3. Refer section 2
4. Refer section 4
5. Refer section 4
6. Refer section 4